



GE Medical Systems

Technical Publications

File Name 105-0127

Revision 1

TEAL PDU MODULE IN RF/PDU CABINET

Copyright[®] 1997-1998 by General Electric Co.

Operating Documentation

TABLE OF CONTENTS

1- INTRODUCTION	2
1-1 Product Description	2
1-2 Siting	4
1-3 Lockout Tagout Procedure	4
1-4 Locking Input Circuit Breaker Procedure	5
2- SET UP & CALIBRATIONS	5
2-1 Connections	5
2-2 Voltage Adjustment	6
2-3 Important Grounding Information.....	9
2-4 Options	10
3- FUNCTIONAL CHECKOUTS.....	10
3-1 Control Circuit Activation.....	10
3-2 Power Off.....	10
3-3 Phase Indicators	11
4- TROUBLESHOOTING	11
4-1 Theory of Operation	11
4-2 Diagnostic Troubleshooting	12
5- FIELD REPLACEMENT UNITS (FRU'S)	13
5-1 Separate Replacement Procedures.....	13
5-1-1 Control Circuit Regulator	13
5-1-2 Control PC Board	14
5-1-3 Fuse Package	14
5-1-4 Suppressor Module	14
5-1-5 Relay Module	15
6- OPTIONAL FEATURES.....	15
6-1 Uninterruptible Power Supply (UPS) Option	15
6-2 Voltage Regulator Option	16
7- DRAWINGS.....	16
7-1 Dimension Diagram.....	17
7-2 Full Component Layout	18
7-3 Block Diagram	19
7-4 Full Schematic	20
7-5 Elementary Diagram	21
7-6 Front View w/o Cover	22
7-7 Left Side	23
7-8 Component Plate	24
7-9 Rear Plate Markings.....	25
7-10 Pin Connections	26
REVISION.....	27

1- INTRODUCTION

1-1 Product Description

Primary function of this unit is distribution and control circuits for powering subsystem equipment of the MR System. The integrated filter in the PDU removes noise and spikes from the AC power line. It is installed between the source of power and the equipment that needs protection from power line disturbances.

The PDU assembly consists of five basic components:

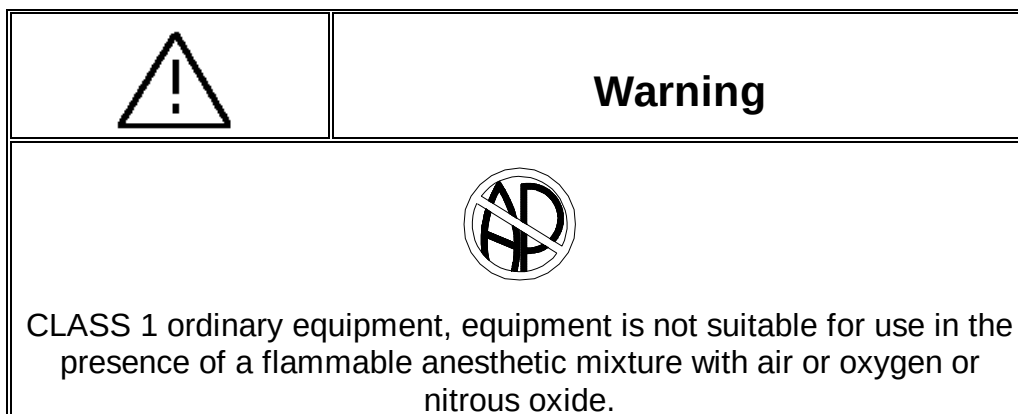
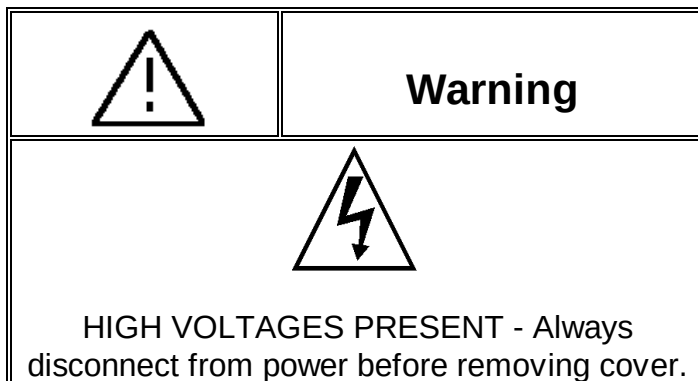
- Distribution and control circuits are included for interface to the MR loads. Provision for connection of optional voltage regulator and UPS are also included, as well as mains adaptation to 380, 400, 415, 480 VAC, 50/60 Hertz.
 - A very well insulated, well shielded, low impedance isolation transformer. This provides protection against "common mode" noise and spikes (neutral-ground potentials). The system can withstand and protect against impulses up to 6000 volts. It can also provide the low duty cycle peak currents required by the MR equipment.
 - A low pass power filter with an exceptional ability to suppress "normal mode" noise (fast line-neutral voltage spikes and dips).
- 
- The image shows a tall, white, modular PDU unit. It has a black frame and a control panel at the bottom. The control panel features a digital display, several buttons, and a row of indicator lights. The unit is standing on a light-colored floor against a wooden wall.
- A modular, field replaceable, three phase surge suppressor, which shunts large impulses. While the isolation transformer and low pass filter handle virtually all disturbances, this surge suppressor's function is to reduce unusually large voltage spikes to levels that are harmless to electronic equipment.
 - Voltage monitor outputs are provided for safer measurement of input and output voltages.

TABLE 1-1
SPECIFICATIONS

COMMON SPECIFICATIONS	ALL MODEL NUMBERS				
<i>Input Voltage</i>	380/400/415/480 Volts DELTA				
<i>Frequency</i>	50/60 Hertz				
<i>Output Voltage</i>	120/208 Volts WYE				
<i>Surge Withstand</i>	Will withstand and protect against worst case power line surges as defined by ANSI/IEEE C62.41 Categories A and B up to 6000V, 3000A capability ringwave and impulse waveform.				
<i>Construction Materials</i>	All transformer wires are insulated with Haptz Class H materials. Windings are insulated with a Nomex paper insulation system before vacuum varnishing. Internal wiring is minimum 200 degree C rated and sleeved with Class H material. A grain-oriented silicon iron core with 14-mil laminations interleaved 1-1 is used to minimize eddy currents and to keep external magnetic fields below 0.1 oersted (measured at 12 inches from unit in any direction).				
INDIVIDUAL SPECIFICATIONS	MODEL NUMBERS				
<i>Model Number</i>	SRFD PDU-1.5				
<i>(TEAL) Part Number</i>	4200010				
<i>Revision</i>	A1				
<i>Power Rating</i>	50 KVA(I)				
<i>Overload Protection (380-415V)</i>	60 Amps				
<i>Overload Protection (480V)</i>	50 Amps				
<i>Load Current</i>	83.3 Amps				
<i>Weight Pounds (Kg.)</i>	800 (362)				
<i>Transfer Impedance at 50 Hz (L- N) 60 Hz (L- N)</i>	< .014 Ω < .015 Ω				
<i>Efficiency Percent</i>	98.4 %				
<i>Load Regulation (0-100% Continuous Load)</i>	$\pm$ 0.6 %				
<i>Load Regulation Peak Load</i>	1.7 %				
<i>Heat Loss (BTU/Hr.)</i>	< 1750				

1-2 Siting

The PDU module is mounted in the lower portion of the RF/PDU cabinet. Refer to the attached mechanical drawing for PDU module specific dimensional information. The unit should be mounted in a dry location. Conduit entrance is through the conduit landing plate at rear of unit.



If the unit is mounted on a raised floor (such as a computer room floor) provisions must be made to support the weight of the unit.

1-3 Lockout Tagout Procedure

DANGER!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN WHEN ALL THE PDU BREAKERS ARE OFF. ENSURE THAT POWER AT MAIN DISCONNECT OR A1 CONTACTOR IS OFF, LOCKED, AND TAGGED.

1. Turn facility circuit breaker (Main Disconnect Control or for Mobile system A1 Contactor) OFF, lock, and tag.

1-4 Locking Input Circuit Breaker Procedure

DANGER!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN WHEN ALL THE PDU BREAKERS ARE OFF.

1. Locate Input Circuit Breaker. Refer to Section 7-6 Front View W/O Cover illustration.
2. Using index finger, push on the arrow symbol located on the breaker handle. This will pop up the locking arm.
3. Place a pad lock on the locking arm.

2-SET UP & CALIBRATIONS

2-1 Connections

The PDU is connected through use of terminal blocks and D-sub connectors at the rear of the unit. Two (2) L21-30R locking receptacles and one (1) 5-15R duplex receptacle are provided at the front of the unit.

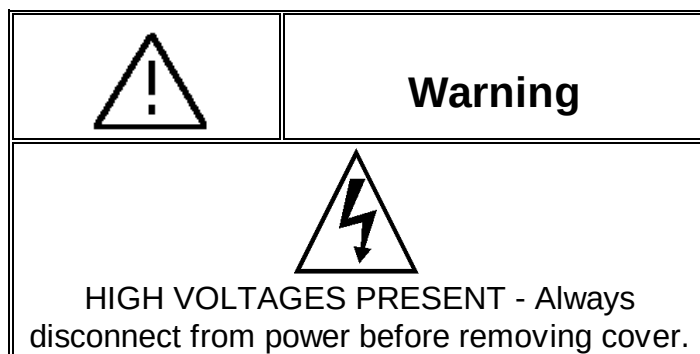
Input and output wires should not be run through the same conduit and should be kept separated if possible. To make power connections to the unit, a conduit landing plate with strain reliefs is provided at the rear of the unit. All neutral connections are to be made to the neutral bar. All ground connections are to be made to the ground bar. Both neutral and ground bars are designed to accommodate two terminations per 1/4-20 stud. All other power connections are made to compression type terminals. The rear connection means are protected by rear covers. The input terminal block is protected from accidental contact by an additional lexan cover. All covers must be installed during normal operation.

Control circuit connections are made using the four D-sub connectors at the rear of the unit.

Check the labels on the PDU to be certain the input and output voltages match the system's requirements. This unit has multiple input voltages which can be changed in the field. There should be a tag attached to the input circuit breaker specifying which input voltage tap has been selected at the factory. Check the voltage present at the site to determine if the PDU input voltage is correctly set. If not, refer to Section 2-2 Voltage Adjustment for instructions.

2-2 Voltage Adjustment

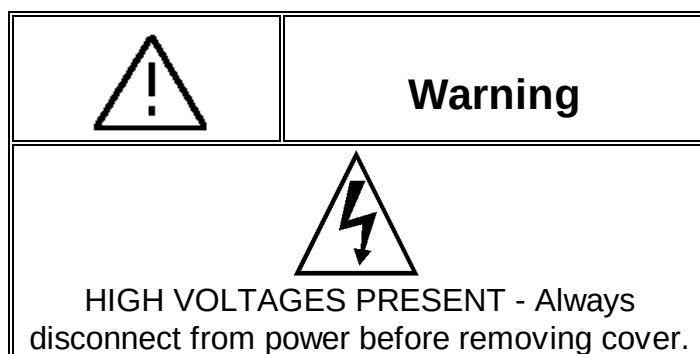
This unit is equipped with voltage adjustment terminals to compensate for voltages in different areas. The available nominal input voltages are 380V, 400V, 415V, and 480V Delta. This adjustment may be done only by qualified personnel as it involves a high voltage electrical connection. Failure to make this adjustment will result in high or low voltage output from the PDU which could potentially harm the loads.



Refer to Section 7-4, Full Schematic and the Voltage Selection Diagram (see below) for information on the available taps. When changing the input voltage setting, it may be necessary to change the amperage setting of the input circuit breaker.

Changing input voltage steps:

1. Lock off the breaker or switch which feeds power to the unit.

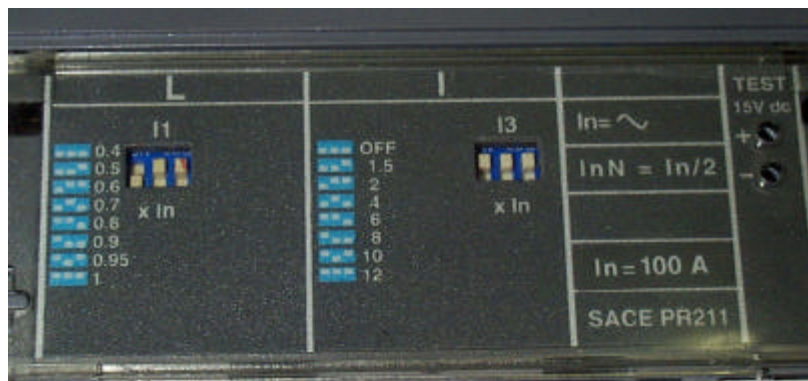


2. Remove lower front cover and set aside.
3. Locate the voltage adjustment terminals. Refer to Section 7-6, Front View w/o Cover diagram.
4. Move the appropriate wires as per the Voltage Selection page (pg. 8) to the desired voltage setting.
5. Verify the input circuit breaker is set correctly. Refer to the table 2-1.

TABLE 2-1
I1 SWITCH SETTINGS FOR INPUT CIRCUIT BREAKER (CB1)

DESIRED INPUT VOLTAGE	380V Delta	400V Delta	415V Delta	480V Delta
CIRCUIT BREAKER AMPERAGE	60 Amps	60 Amps	60 Amps	50 Amps
CIRCUIT BREAKER "L" SETTING	0.6	0.6	0.6	0.5
SWITCH SETTING	___ --- ---	___ --- ---	___ --- ---	___ ___ ---

I3 SETTING: Leave I3 setting at factory set condition of 12 x In



6. If input breaker setting does not need to be changed, skip to step 9, otherwise continue to step 7.
7. Locate the "L" setting section on the face of the input circuit breaker.
8. With a small blade screwdriver move the dip switch toggles to the appropriate setting as per the above table.
9. Ensure all connections are tight and secure front cover.
10. With all circuit breakers in the off position, apply voltage to the unit.
11. Turn on the input circuit breaker and energize the control circuit.
12. Check both input and output voltages at the test points on the voltage divider board. The measured voltage will be 1/100th of the actual voltage ($\pm 1\%$).
13. If voltages check out, turn the loads on individually.

VOLTAGE SELECTION

Wired for 380V

TB2-TB16

O-A

380-A

400-A

415-A

480-A

O-B

380-B

400-B

415-B

480-B

O-C

380-C

400-C

415-C

480-C

Wired for 400V

TB2-TB16

O-A

380-A

400-A

415-A

480-A

O-B

380-B

400-B

415-B

480-B

O-C

380-C

400-C

415-C

480-C

Wired for 415V

TB2-TB16

O-A

380-A

400-A

415-A

480-A

O-B

380-B

400-B

415-B

480-B

O-C

380-C

400-C

415-C

480-C

Wired for 480V

TB2-TB16

O-A

380-A

400-A

415-A

480-A

O-B

380-B

400-B

415-B

480-B

O-C

380-C

400-C

415-C

480-C

WARNING! HIGH VOLTAGES
PRESENT INSIDE UNIT, TO BE
OPENED ONLY BY QUALIFIED
PERSONNEL

AD 10-30-98 NEW RELEASE 10-30-98

TEAL

TEAL ELECTRONICS CORPORATION
SAN DIEGO, CALIFORNIA

DATE 10-29-98

TECHNICALY APPROVED BY *[Signature]*

RELAXED *[Signature]*

50 KVA() SRFD PDU-1.5
POWER CONDITIONER
VOLTAGE SELECTION

MODEL 153-127

SCALE (1000 WATT) 1050127

THIS DOCUMENT CONTAINS CONFIDENTIAL, PROPRIETARY,
INFORMATION AND IS NOT TO BE DISTRIBUTED OUTSIDE
TEAL ELECTRONICS CORPORATION.

REVISION 10-30-98

APPROVED *[Signature]*

TECHNICALY APPROVED BY *[Signature]*

RELAXED *[Signature]*

105-0127

V-1

8

2-3 Important Grounding Information

Good grounding is critical to the reliable operation of electronic systems as well as personnel safety. The following directions should be followed only in as far as they are consistent with all local and national electrical codes. Grounding can often be confusing and difficult. Contact TEAL Electronics at 619-558-9000 if you wish more specific advice.

This PDU includes an isolation transformer that can significantly improve the quality of system grounding. By re-referencing the phase conductors to ground, it eliminates incoming line-side "common mode" noise (and complies with Article 250-5 of the NEC). The output neutral (if present) is tied to ground.

Input service ground requirements: If all communicating system components are powered from this single point ground with isolated ground-type distribution, the system is then completely isolated from external ground quality issues. Non-isolated code minimum input grounds can then be used. Verify input grounding meets codes prior to installation.

If the system has non-isolated communications lines from other devices not powered by the PDU, the system is then less isolated and incoming ground quality becomes more important. The communications line becomes a "back door" through which power and grounding problems can disrupt or damage the system. Good grounding of devices on both ends of the line helps minimize potential problems, as would surge suppressors on the communications line or a PDU on the device on the other end. Surge suppressors on the device, however, hurt much more than they help. Isolation of the communications line (optical or magnetic) eliminates this problem altogether.

If a "good" input ground is required, it should be a continuous, insulated, isolated, oversized wire back to the service entrance. The Federal Information Processing Standards Publication 94 (9/21/83) also known as "FIPS Pub 94" is an excellent reference guide. It defines a "good ground" as one with low impedance over a wide spectrum of high frequencies. It also says that optimum equipment performance and personnel safety can be achieved by tying all communicating system components to a single point ground (or, in the case of a large room, create a "ground reference plane" that acts as a single point).

Local Grounding: Article 250-26 of the National Electric Code (NEC) requires that hard wired PDUs over 1 KVA must also be bonded to nearest available local building ground (typically building steel) as well as the incoming service ground. This also helps by referencing the system to local ground potential in case parts of the system might be touching grounded parts. We recommend against the use of special dedicated ground rods, as they are expensive, usually don't improve and often hurt grounding quality, can create safety hazards and in most cases may violate NEC 250-54.

Isolated output grounding: Except at the point of common connection (PCC) at the PDU, it is important that the output ground from the PDU be kept isolated from building ground in any output wiring. If receptacles are hard wired to the PDU in the field, isolated ground receptacles must be used. These special receptacles avoid re-connecting the output ground to the conduit, and because they look different they reduce the chance of non-critical loads being inadvertently plugged into the conditioned power.

2-4 Options

Refer to the appropriate section of option information.

- Section 6-1 Uninterruptible Power Supply
- Section 6-2 Regulator

3- FUNCTIONAL CHECKOUTS

Once voltage has been applied to the unit, without a load connected, turn on the main input circuit breaker. Energize the control circuit by placing a jumper across pins J2-7 and J2-8 and then push the Emergency Off (EMO) reset button. Check both input and output voltages at the test points on the voltage divider board. The measured voltages will be 1/100th of the actual voltage ($\pm 1\%$).

If the voltages check out, connect the loads and turn on individually. If the PDU is overloaded, the input circuit breaker will shut off. Otherwise, everything should operate normally. If voltages do not check out, refer to section 4-Troubleshooting.

The following functions should be verified during installation. All functions (not just the ones mentioned here) were completely tested at the factory and found to be working properly. These functional checks should still be performed upon installation as a failure of any one function could create an unsafe condition.

3-1 Control Circuit Activation

Initial condition:	Power applied, input circuit breaker ON, and a jumper placed across pins J2-7 and J2-8.
Action:	EMO reset switch pushed.
Response:	Contactor coils should energize and all functions of the unit should be working. This is the normal operation state.

3-2 Power Off

Initial condition:	Power applied, input circuit breaker ON, and control circuit activated as per above.
Action:	Red, Power Off switch pushed.
Response:	Input circuit breaker should trip.

3-3 Phase Indicators

Initial condition: Power applied, input circuit breaker ON, and control circuit activated as per above.

Action: Visually inspect that all phase indicators are lit. If any indicator is not lit, refer to section 4- Troubleshooting.

Response: N/A

4- TROUBLESHOOTING

4-1 Theory of Operation

The unit is designed to supply 120/208V WYE to numerous uncontrolled outputs and four (4) controlled outputs. The SGD1, SGD2, Resistive Shim, and RF Amp/Pen outputs can be turned off remotely in the event an emergency arises and as needed for servicing subsystems at the MR System. These outputs use 120V contactors (both coil and contacts) switched with a relay using a 24VDC coil. This 24VDC circuit is called the Control Circuit.

The Control Circuit consists of a 120/24VAC step down transformer supplying a regulator to convert to 24VDC which supplies a printed circuit board called the Control PC Board. The Control PC Board consists of relays, output D-Sub connectors, and interconnection connectors.

The Control Circuit is the primary means by which the input circuit breaker can be tripped in the event of an emergency or internal failure. Since tripping the input breaker requires a manual reset, the four (4) controlled outputs can be turned off remotely without the need to reset the input breaker. The control circuit will still need to be re-energized via the Emergency Off (EMO) reset switch.

In addition, the unit is also supplied with a Voltage Divider PC Board. This PC board is designed to allow safe testing of the input and output voltages. The board supplies test points that are 1/100th ($\pm 1\%$) of the actual voltages. This board also has phase indicators to provide basic feed back in the event a problem exists with the supplied phases.

4-2 Diagnostic Troubleshooting

TABLE 4-1
DIAGNOSTIC TROUBLESHOOTING

Condition	Possible Cause	Solution
Input circuit breaker tripped	- Power Off switch was activated - Thermal cutout closed (transformer too hot) - Suppressor module needs replacement - Estop switch activated - Stop switch activated - Humidity contact closed	- Correct condition and turn on input circuit breaker - Requires service call to factory (various causes) - Replace defective module - Correct condition and turn on input breaker - Correct condition and turn on input breaker - Correct condition and turn on input breaker
Output circuit breaker tripped (any)	Overload condition for respective circuit	Correct overload condition
No output at SGD 1, SGD 2, RF Amp/Pen, Resistive Shim (if option is used)	- Emergency Off switch activated - EMO reset switch not pushed on startup - Defective contactor coil or relay - Input circuit breaker tripped	- Correct emergency condition and push EMO reset - Energize control circuit by pushing EMO reset switch - Replace contactor coil or relay - Refer to Input circuit breaker tripped condition, above
Control circuit not working (EMO reset activated)	- Fuse F1 blown - Fuse F2 blown - REG1 Failure	- Find electrical short or correct overload cause and replace fuse - Refer to 5-1-1, Control Circuit Regulator (REG1)
Output voltage too high or too low	Improper input taps selected	Select proper input taps
Phase indicator(s) not lit	- Loss of power at respective phase - Indicator defective	- Correct condition - Replace Voltage Divider PC Board

5- FIELD REPLACEABLE UNITS (FRU'S)

TABLE 5-1
FIELD REPLACEMENT UNITS

DESCRIPTION	FRU TYPE	REFERENCE	MFG. PART NO.	GE PART NO.
Control Circuit Regulator	FRU1	REG1	4990259	2185383-2
Control PC board	FRU1	PCB2	4990260	2185383-3
Emergency reset switch	FRU2	SW2	4990261	2185383-4
Filter Capacitor, main	FRU2	C1 - C3	4990262	2185383-5
Filter capacitor, snubber	FRU2	C4 - C6	4990263	2185383-6
Fuse package	FRU1	F1	4990264	2185383-7
Fuse Package	FRU1	F2	4990264	2185383-7
Input Circuit Breaker shunt trip module	FRU2	ST	4990266	2185383-10
Suppressor module	FRU1	Q1 - Q3	4990267	2185383-11
Power Off switch	FRU2	SW1	4990268	2185383-12
Relay module	FRU1	RY1	4990269	2185383-13
Voltage divider PC board	FRU2	PCB1	4990270	2185383-14
Tap selection jumper wires	FRU2	---	4990271	2185383-15
Circuit breaker 63A	FRU2	CB2 - CB3	4990273	2185383-16
Circuit breaker 32A	FRU2	CB5, CB11, CB12, CB14	4990274	2185383-17
Circuit breaker 25A	FRU2	CB6	4990275	2185383-18
Circuit breaker 20A	FRU2	CB8	4990276	2185383-19
Circuit breaker 16A, 2-Pole	FRU2	CB10	4990277	2185383-20
Circuit breaker 16A, 1-Pole	FRU2	CB7, CB9, CB13	4990272	2185383-8
Circuit breaker 40A	FRU2	CB4	0450496	2224561-2

5-1 FRU1 Replacement Procedures

5-1-1 Control Circuit Regulator (REG1)

Note: Prior to replacing Regulator Assembly, verify that replacement is required. To do this, verify that voltage at Control Circuit Board pins J4-5 and J4-6 is not present during normal operation.

1. Lock out and tag out the feeder breaker to the unit.
2. Remove rear upper cover and set aside.
3. Locate the Regulator Assembly. Refer to Section 7-8 Component Plate drawing, REG1 is Regulator Assembly.
4. Capacitor will not normally require any manual discharging. As always, safe handling of capacitors should be used.
5. Verify markings of wires connected to the top of the Regulator Assembly.
6. Remove the wires connected to the top of the Regulator Assembly.
7. Loosen the clamp around the capacitor of the Regulator Assembly and remove the Regulator Assembly.
8. Replace with new Regulator Assembly.
9. Reconnect wires to Regulator Assembly removed in Step 5.

1. Tighten clamp around the capacitor of the Regulator Assembly.
2. Reinstall both front covers.
3. Turn on the main input breaker.
4. Push the Emergency Off (EMO) reset button.
5. Verify voltage at Control Circuit Board pins J4-5 and J4-6 is 24 VDC.

5-1-2 Control PC Board (PCB2)

1. Lock off input circuit breaker CB1.
2. Locate Control PC Board. Refer to Section 7-7 Left Side illustration.
3. Disconnect any connectors connected to J1 through J4 on Rear Upper Panel.
4. Remove rear upper cover and set aside.
5. Remove wire harnesses connected to the Control PC Board.
6. Locate and remove the Control PC Board from the four (4) standoffs. Refer to Left Side page.
7. Replace with new Control PC Board.
8. Reconnect wire harnesses.
9. Reinstall rear upper cover.
10. Reconnect any connectors removed in Step 2.
11. Remove lock out from input circuit breaker CB1 previously installed in Step 1.
12. Turn on input breaker CB1.
13. Push the Emergency Off (EMO) reset button.

5-1-3 Fuse Package (F1 or F2)

1. Turn off input circuit breaker CB1.
2. Locate fuse on top front cover. Refer to Section 7-7 Left Side illustration.
3. Twist off fuse cover and remove defective fuse.
4. Replace with fuse of same size and rating.
5. Turn on input circuit breaker CB1.
6. Push the Emergency Off (EMO) reset button.
7. Verify voltage at Control Circuit Board pins J4-5 and J4-6 is 24 VDC.

5-1-4 Suppressor Module (Q1 - Q3)

1. Locate Suppressor Modules on front of PDU. Refer to Section 7-7 Left Side illustration.
2. Remove Suppressor Cover plate and set aside.
3. Remove defective Suppressor Module by firmly pulling out on the failed module.
4. Replace with new module.
5. Reinstall Suppressor Cover plate.
6. Turn on input circuit breaker CB1.
7. Mechanical failure flag on Suppressor Module should not be visible.

5-1-5 Relay Module (RY1)

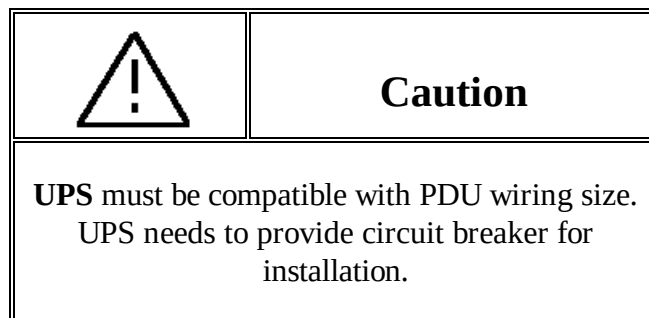
1. Turn off input circuit breaker CB1.
2. Remove top rear cover.
3. Locate Relay Module. Refer to Section 7-8 Component Plate illustration.
4. Slide hold down clamp off top of Relay Module.
5. Unplug Relay Module from base.
6. Replace with new Relay Module. Relay Module can only be installed in one direction. Do not use excessive force.
7. Slide hold down clamp over top of Relay Module.
8. Reinstall top rear cover.
9. Turn on input circuit breaker.
10. Push the Emergency Off (EMO) reset button.
11. Verify voltage at Control Circuit Board pins J4-5 and J4-6 is 24 VDC.

6- Optional Features

6-1 Uninterruptible Power Supply (UPS) Option

A UPS can be installed to the terminals at the rear of the unit. Conduit landing locations are provided with strain reliefs. Refer to the Rear Plate Markings page for location of UPS terminals.

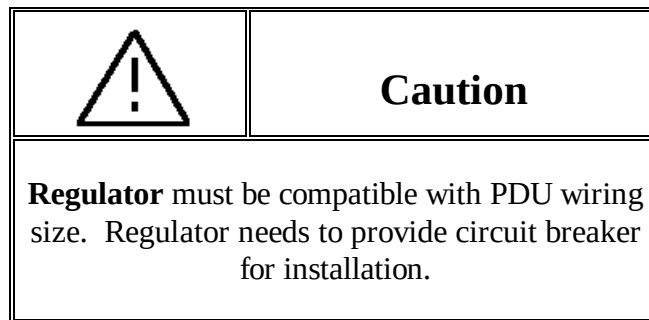
If a UPS is installed and building or system power is lost, the UPS will continue to supply power to the System Cabinet, and Operator Workspace (PC and Host) loads.



6-2 Voltage Regulator Option

A Teal Voltage Regulator may be installed to the terminals at the rear of the unit. Conduit landing locations are provided with strain reliefs. Refer to the Rear Plate Markings page for location of Regulator Input and Output terminals.

The Regulator Input terminals are to be used for connecting the input wires of the regulator. While the Regulator Output terminals are to be used for connecting the output wires of the regulator. Refer to the installation instructions for the regulator for more detailed installation instructions.



7- DRAWINGS

7-1 Dimension Diagram

7-2 Full Component Layout

7-3 Block Diagram

7-4 Full Schematic

7-5 Elementary Diagram

7-6 Front View w/o Cover

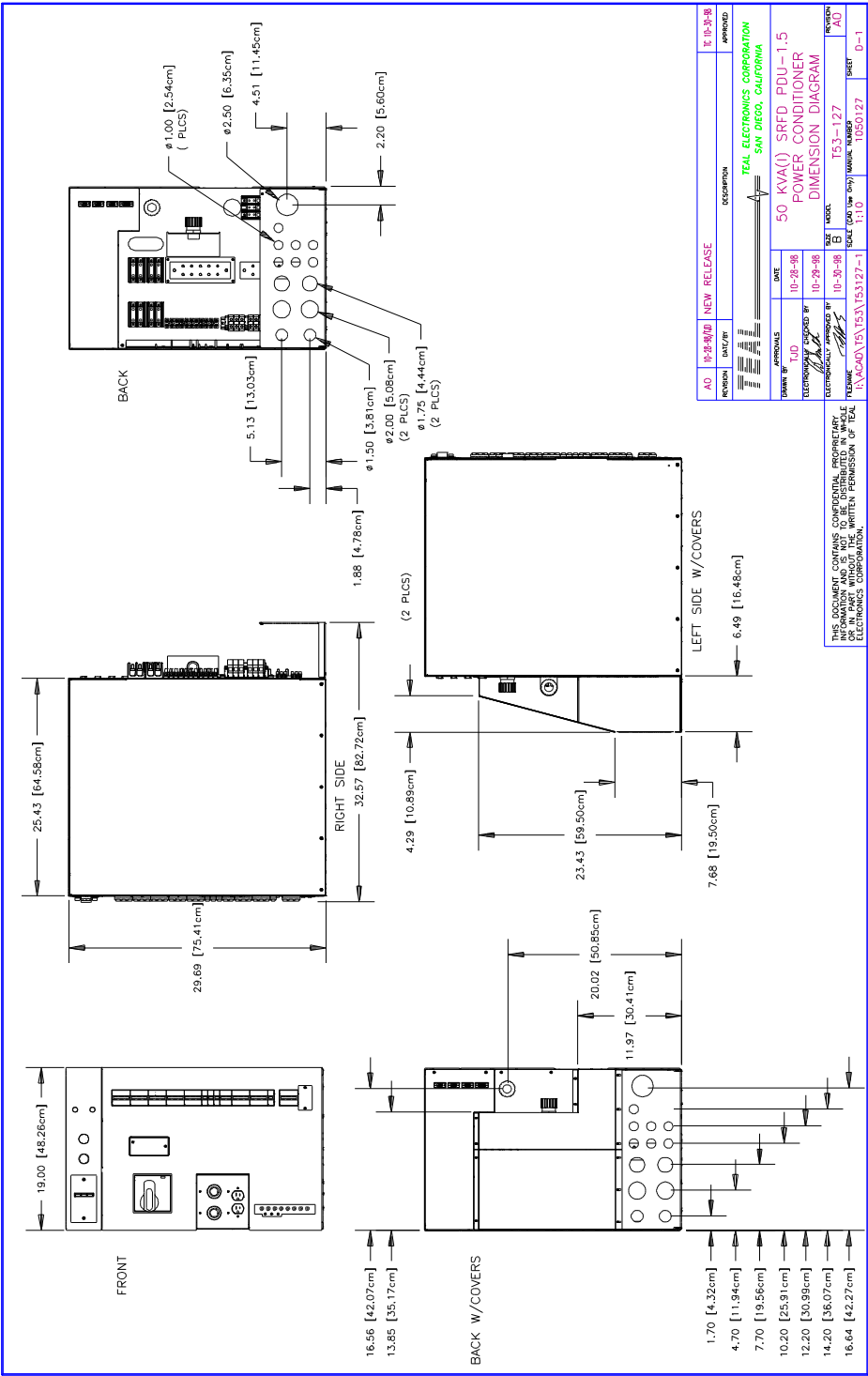
7-7 Left Side

7-8 Component Plate

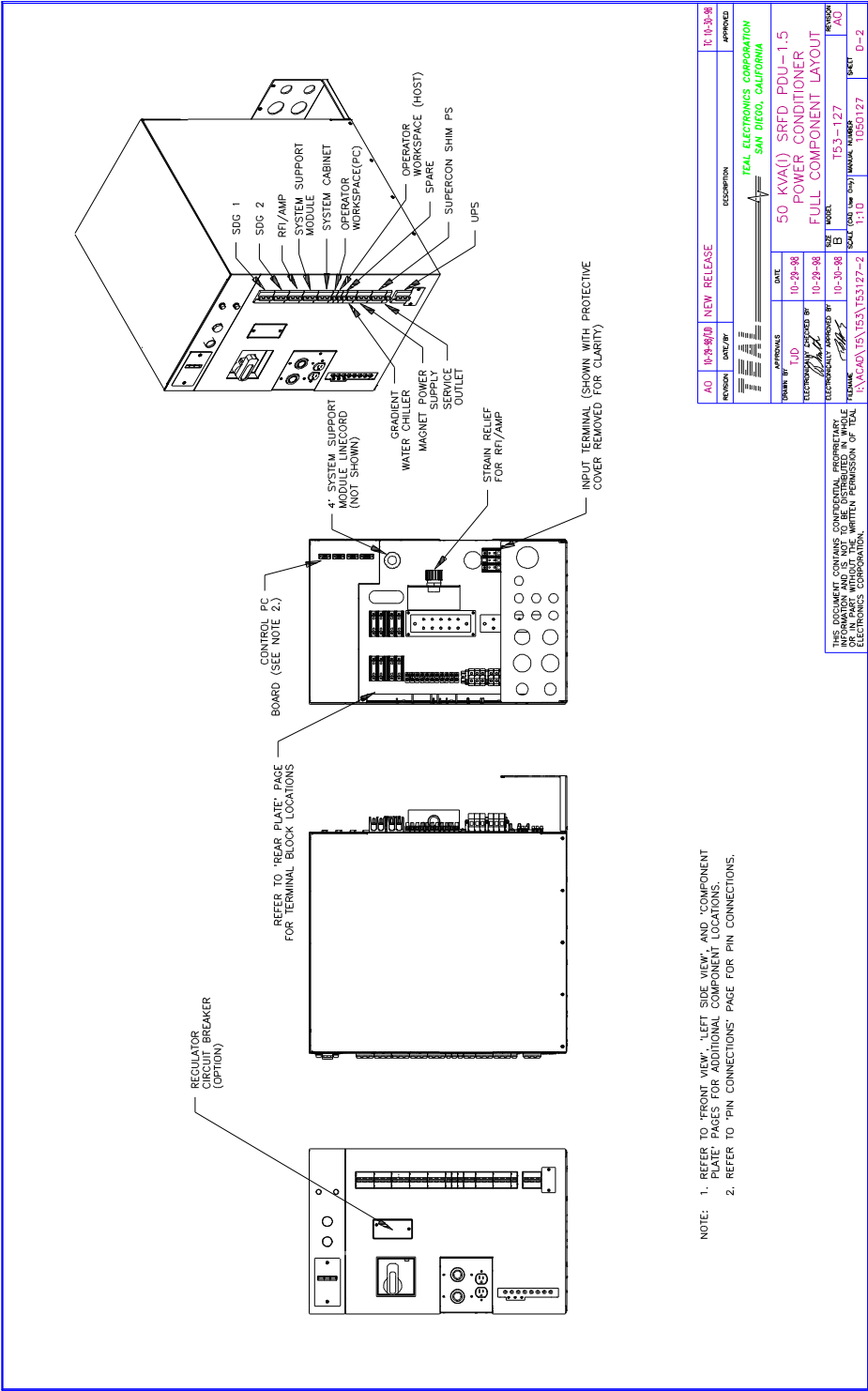
7-9 Rear Plate Markings

7-10 Pin Connections

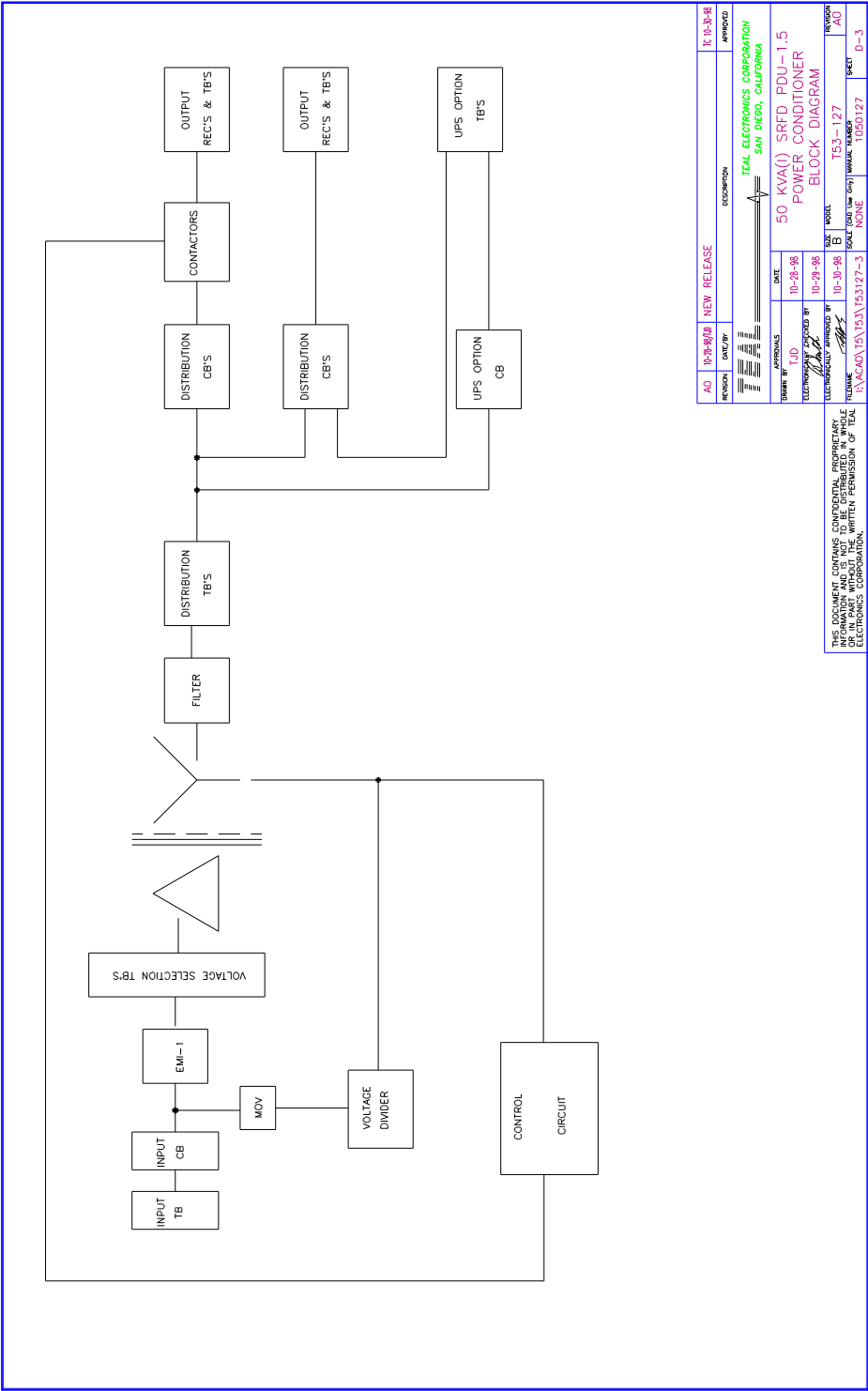
7-1 DIMENSION DIAGRAM



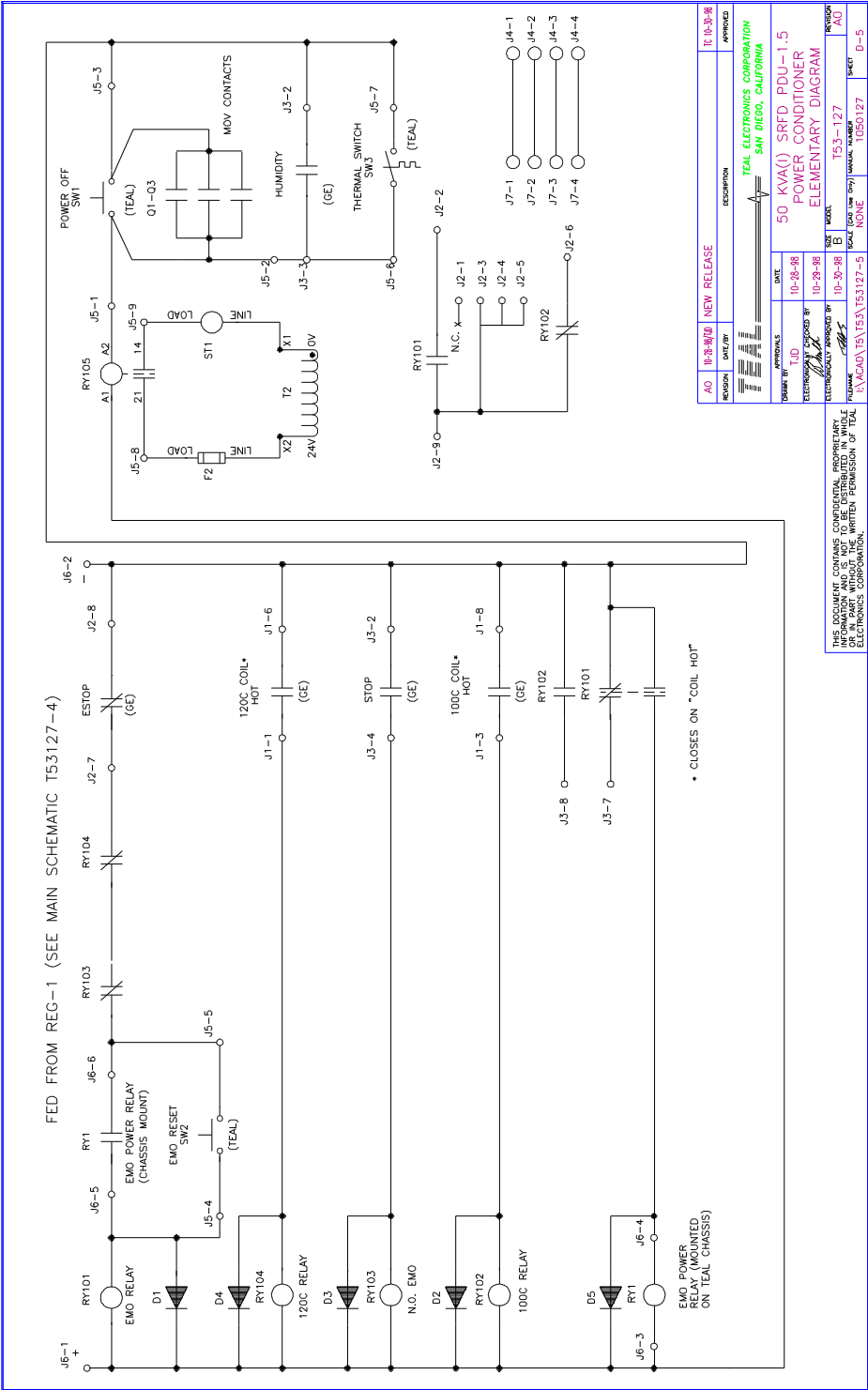
7-2 FULL COMPONENT LAYOUT



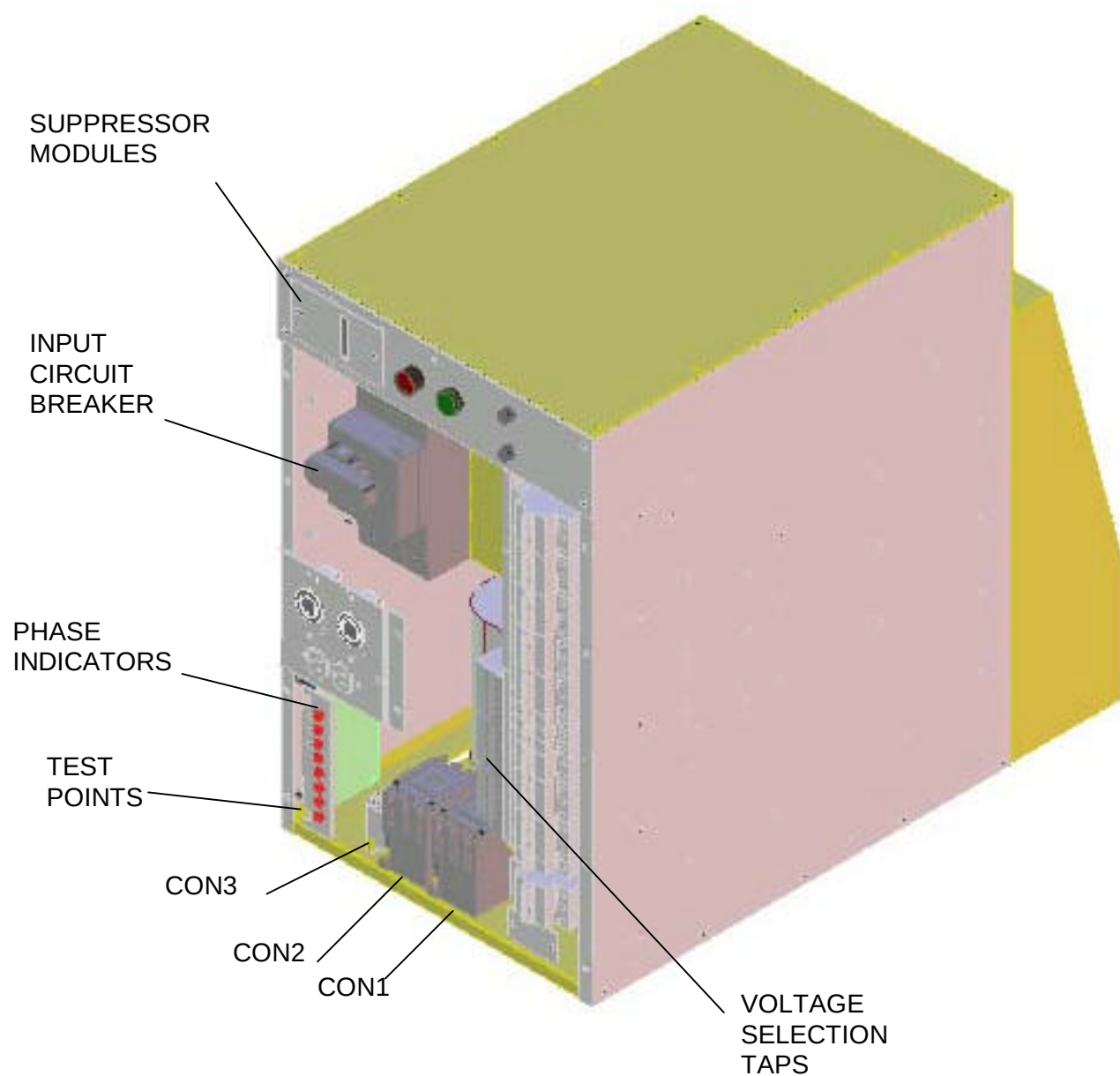
7-3 BLOCK DIAGRAM

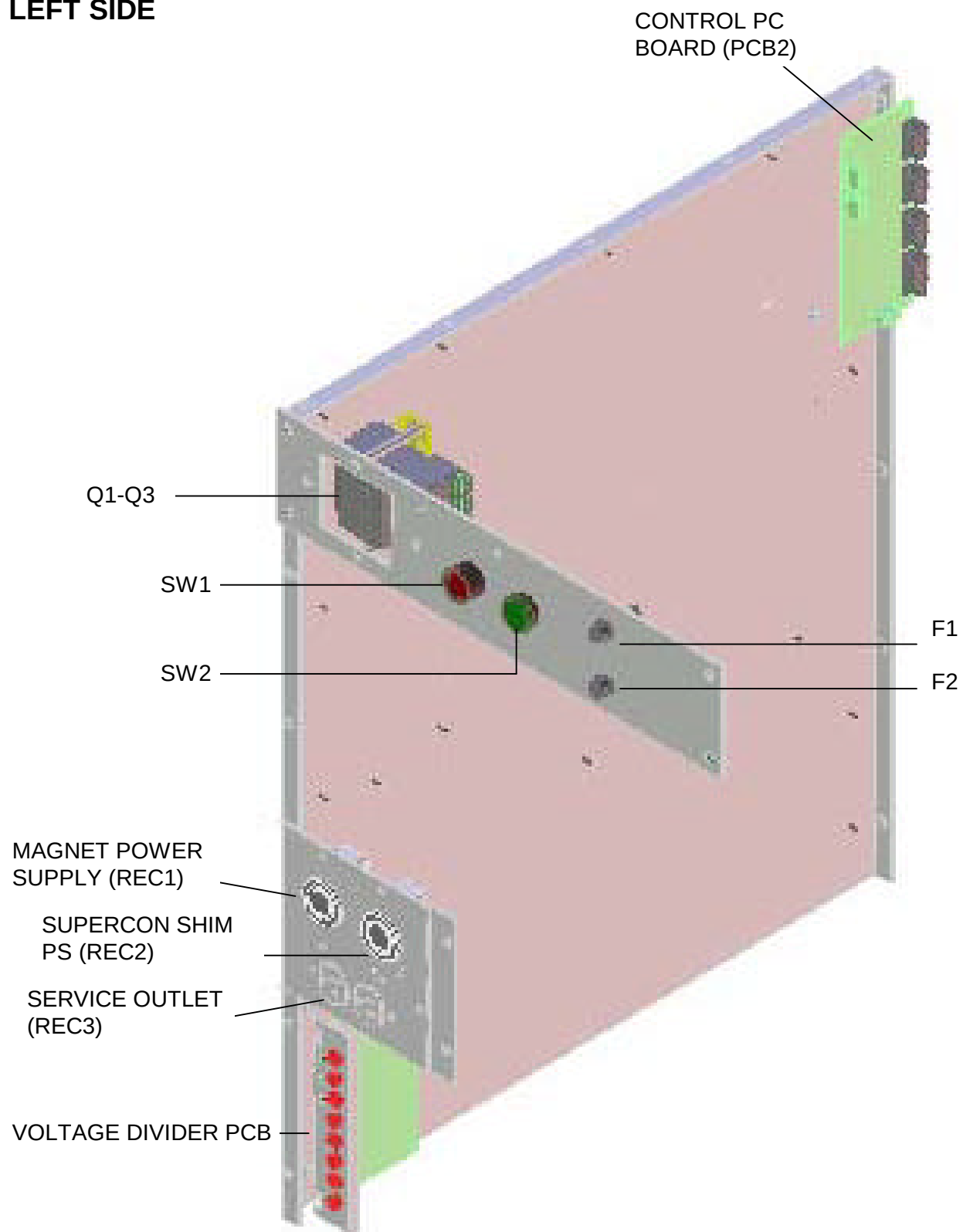


7-5 ELEMENTARY DIAGRAM

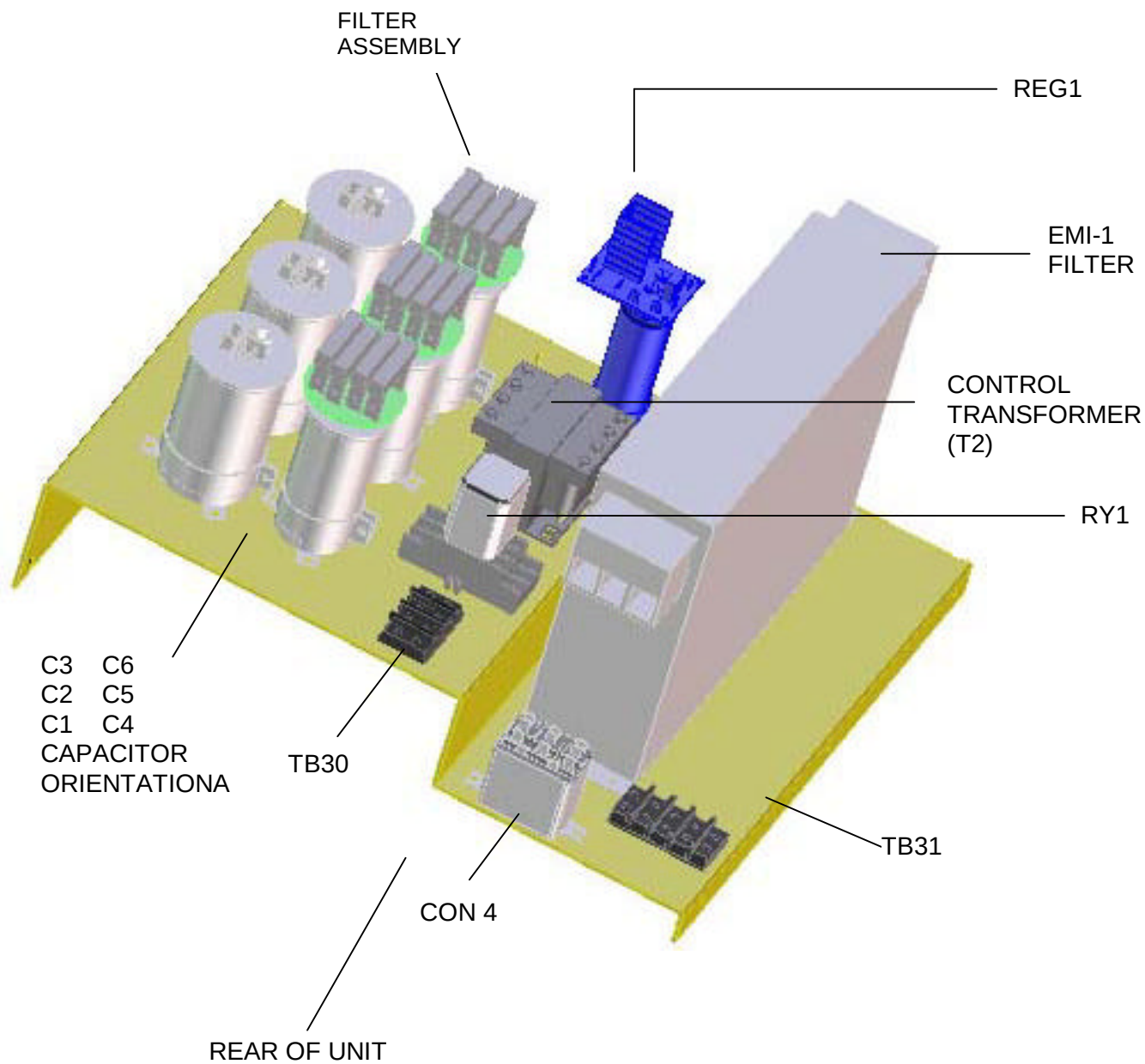


7-6 FRONT VIEW W/O COVER

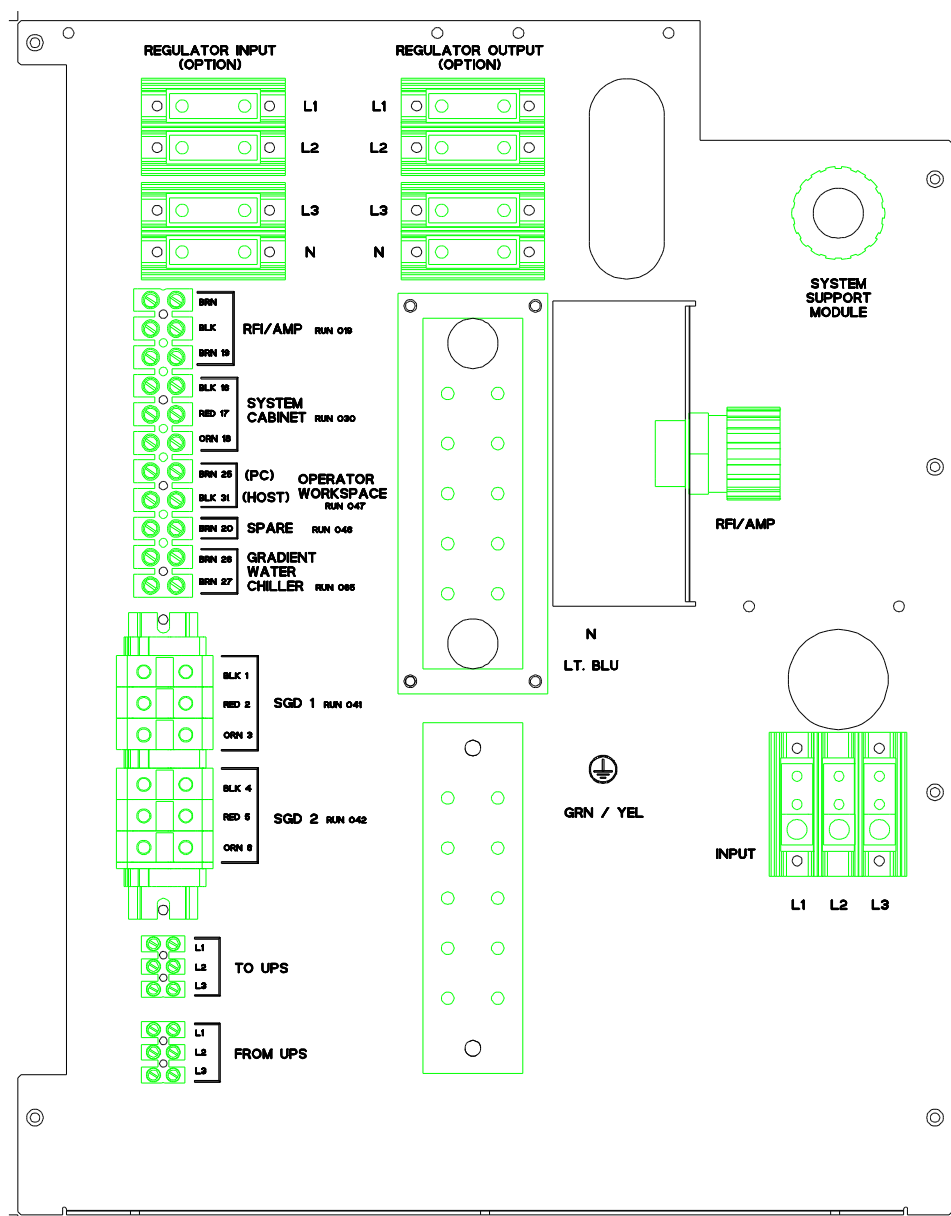


7-7 LEFT SIDE

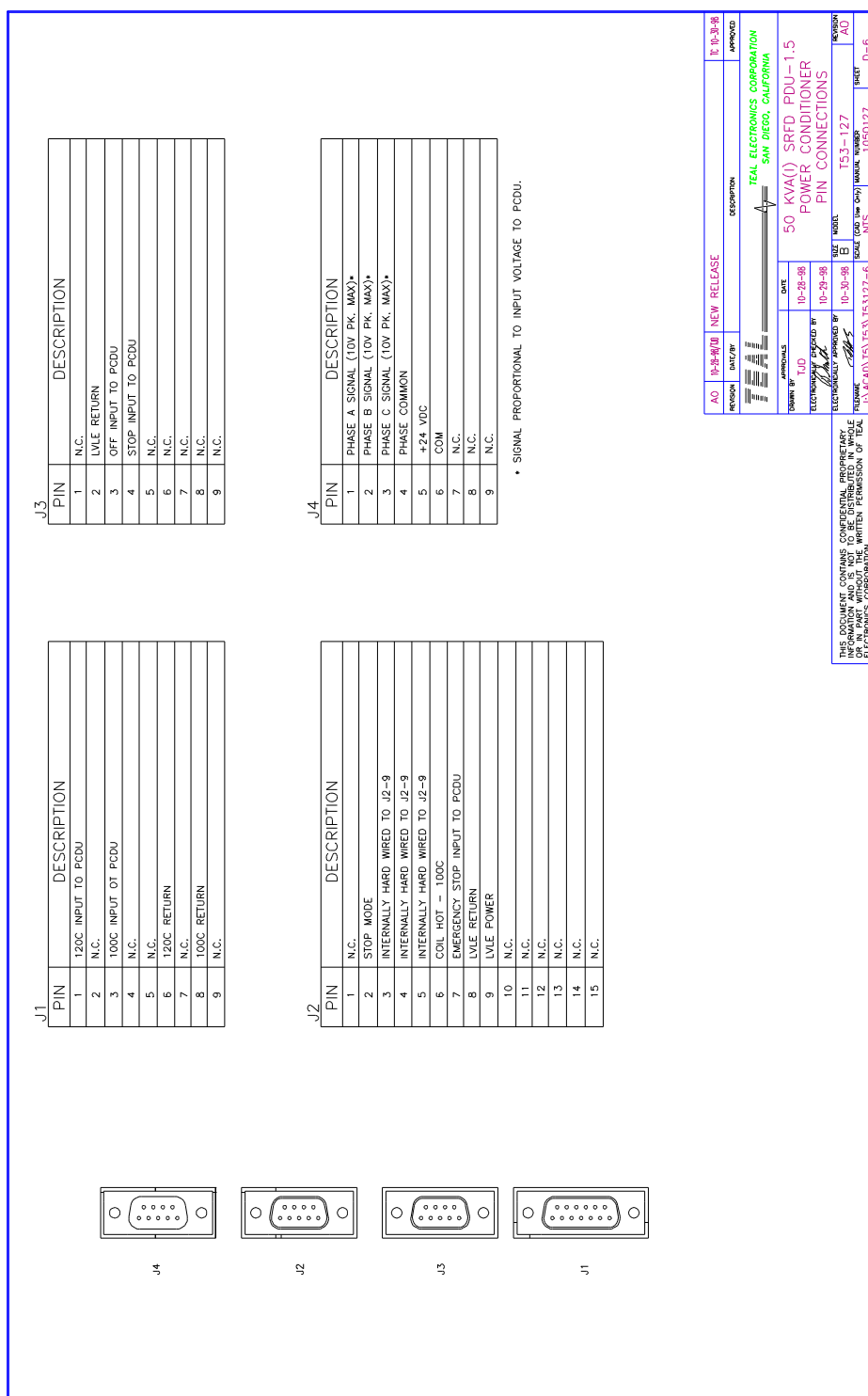
7-8 COMPONENT PLATE



7-9 REAR PLATE MARKINGS



7-10 PIN CONNECTIONS



REVISION

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	October 16, 1998	T. Donaldson	Initial release.
1	April 19, 1999	T. Donaldson	RF1/AMP Circuit Previously 32 Amp, now 40 Amp.